

Electrodynamics and Relativity

Problem 12.1

Problem 12.2

Problem 12.3

Problem 12.4

As the outlaws escape in their getaway car, which goes $\frac{3}{4}c$, the police officer fires a bullet from the pursuit car, which only goes $\frac{1}{2}c$. The muzzle velocity of the bullet (relative to the gun) is $\frac{1}{3}c$. Does the bullet reach its target (a) according to Galileo, (b) according to Einstein?

Solution:

(a) Adding up the velocities with Galileo's velocity addition rule, the resulting velocity of the bullet is

$$\frac{1}{2}c + \frac{1}{3}c = \frac{5}{6}c > \frac{3}{4}c$$

so the bullet does reach its target.

(b) Using Einstein's velocity addition rule, the result is instead

$$v_{\text{bullet}} = \frac{v_{\text{car}} + v_{\text{muzzle}}}{1 + \frac{v_{\text{car}} v_{\text{muzzle}}}{c^2}} = \frac{\frac{5}{6}c}{1 + \frac{1}{6}} = \frac{5}{7}c < \frac{3}{4}c$$

so the bullet does not reach its target. $\square$

Problem 12.5**Problem 12.6****Problem 12.7**

In a laboratory experiment, a muon is observed to travel 800 m before disintegrating. A graduate student looks up the lifetime of a muon ($2 \times 10^{-6}\text{s}$) and concludes that its speed was

$$v = \frac{800 \text{ m}}{2 \times 10^{-6}\text{s}} = 4 \times 10^8 \text{ m/s}.$$

Faster than light! Identify the student's error, and find the actual speed of this muon.

Solution:

The student did not take into account the lifetime of the muon in the muons reference time. The muons lifetime in its own reference time is

$$\gamma t = \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} t$$

If it travels for 800m then its speed is

$$\frac{800 \text{ m}}{\gamma \times 2 \times 10^{-6}\text{s}} = v$$

Simplifying the distance over time gives

$$\frac{800 \text{ m}}{2 \times 10^{-6}\text{s}} = \frac{4}{3}c$$

Solving for v

$$\begin{aligned}
\frac{4}{3}c &= \gamma v = \frac{v}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} \\
&\downarrow \\
\left(\frac{4}{3}c\right)^2 &= \frac{v^2}{1 - \left(\frac{v}{c}\right)^2} \\
&\downarrow \\
\left(\frac{4}{3}c\right)^2 \left(1 - \left(\frac{v}{c}\right)^2\right) &= v^2 \\
&\downarrow \\
\left(\frac{4}{3}c\right)^2 &= v^2 \left(1 + \left(\frac{4}{3}c\right)^2 \frac{1}{c^2}\right) = v^2 \left(1 + \frac{16}{9}\right) \\
&\downarrow \\
v &= \frac{4}{3}c \frac{1}{\sqrt{1 + \frac{16}{9}}} = \frac{4}{3}c \frac{1}{\sqrt{\frac{25}{9}}} = \frac{12}{15}c = \boxed{\frac{4}{5}c}
\end{aligned}$$

□

Problem 12.8

Problem 12.9

A Lincoln Continental is twice as long as a VW Beetle, when they are at rest. As the Continental overtakes the VW, going through a speed trap, a (stationary) policeman observes that they both have the same length. The VW is going at half the speed of light. How fast is the Lincoln going?

Solution:

The length of the cars while stationary is denoted L . With length contraction due to the velocity accounted for the length is denoted L' . With v_{vw} given and $L_{\text{lc}} = 2L_{\text{vw}}$, the solution is obtained by setting $L'_{\text{lc}} = L'_{\text{vw}}$ and solving for v_{lc} .

$$L_{lc} = 2L_{vw}$$

$$L'_{vw} = L_{vw} \sqrt{1 - \frac{v_{vw}^2}{c^2}} = L_{vw} \frac{\sqrt{3}}{2}$$

$$L'_{vw} = L'_{lc} = L_{lc} \sqrt{1 - \frac{v_{lc}^2}{c^2}} = L_{vw} \frac{\sqrt{3}}{2}$$

Using $L_{lc} = 2L_{vw}$ and solving for v_{lc} , the velocity for which the length of the Lincoln Continental equals the length of the VW Beetle is

$$v_{lc} = \frac{\sqrt{13}}{4}c$$

□

Problem 12.10

Problem 12.11

Problem 12.12

Solve

- (i) $\bar{x} = \gamma(x - vt)$
- (ii) $\bar{y} = y$
- (iii) $\bar{z} = z$
- (iv) $\bar{t} = \gamma\left(t - \frac{v}{c^2}x\right)$

for x, y, z, t in terms of $\bar{x}, \bar{y}, \bar{z}, \bar{t}$, and check that you recover

- (i') $x = \gamma(\bar{x} + v\bar{t})$
- (ii') $y = \bar{y}$
- (iii') $z = \bar{z}$
- (iv') $t = \gamma\left(\bar{t} + \frac{v}{c^2}\bar{x}\right)$

Solution:

From (i) and (iv) we have

$$(i) \quad \bar{x} = \gamma(x - vt) \rightarrow x = \frac{\bar{x}}{\gamma} + vt$$

$$(iv) \quad \bar{t} = \gamma\left(t - \frac{v}{c^2}x\right) \rightarrow t = \frac{\bar{t}}{\gamma} + \frac{v}{c^2}x$$

Inserting t from (iv) into the expression for x in (i) gives

$$x = \frac{\bar{x}}{\gamma} + v\left(\frac{\bar{t}}{\gamma} + \frac{v}{c^2}x\right) \rightarrow x\left(1 - \frac{v^2}{c^2}\right) = \frac{\bar{x}}{\gamma} + \frac{v\bar{t}}{\gamma}$$

↓

$$x \frac{1}{\gamma^2} = \frac{1}{\gamma}(\bar{x} + v\bar{t}) \rightarrow x = \gamma(\bar{x} + v\bar{t}) = (i')$$

Putting this expression for x into the expression for t in (iv) gives

$$t = \frac{\bar{t}}{\gamma} + \frac{v}{c^2}(\gamma(\bar{x} + v\bar{t})) = \gamma\left(\frac{\bar{t}}{\gamma^2} + \frac{v}{c^2}\bar{x} + \frac{v^2}{c^2}\bar{t}\right) = \gamma\left(\left(1 - \frac{v^2}{c^2} + \frac{v^2}{c^2}\right)\bar{t} + \frac{v}{c^2}\bar{x}\right)$$

↓

$$t = \gamma\left(\bar{t} + \frac{v}{c^2}\bar{x}\right) = (iv')$$

(ii) $\leftrightarrow$ (ii') and (iii) $\leftrightarrow$ (iii') follow directly from the equations.

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